

ExamEasy KENYA - STUDENT RESOURCE PORTAL

CONFIDENTIAL & VERIFIED ACADEMIC RESOURCE

FACULTY: SCHOOL OF COMPUTING & INFORMATICS

COURSE UNIT: BCS 201 - Data Structures and Algorithms

EXAMINATION TYPE: PAST PAPER

ACADEMIC YEAR: 2023 | SEMESTER 1

PRICE: KES 200.00 (PAID & VERIFIED)

SOLUTIONS: COMPREHENSIVE MARKING SCHEME INCLUDED

TOTAL PAGES: 5 PAGES | WATERMARK: EXAMEASY GENUINE COPY

INSTRUCTIONS TO CANDIDATES:

- 1. Answer Question ONE in Section A (Compulsory) and any other TWO questions in Section B.
- 2. Write your answers legibly in the provided answer booklet.
- 3. Non-programmable scientific calculators are permitted.
- 4. Time Allowed: 2 Hours 30 Minutes.

SECTION A: COMPULSORY (30 MARKS)

- 1. (a) Write a C++ or Java method to detect a cycle in a singly linked list using Floyd's Cycle Detection [8 Marks]
- 1. (b) Show that any comparison-based sorting algorithm requires at least $\Omega(n \log n)$ comparisons in the worst case [6 Marks]

SECTION B: ANSWER ANY TWO QUESTIONS (20 MARKS EACH)

- 2. (a) Formulate the optimization function and solve for boundary constraints. [10 Marks]
- 2. (b) Provide step-by-step analytical derivation with full annotations. [10 Marks]
- 3. (a) Evaluate real-world case studies demonstrating strategic failure and remediation. [12 Marks]
- 3. (b) Propose architectural best practices for scalable deployment. [8 Marks]

DETAILED MARKING SCHEME & MODEL SOLUTIONS:

- Question 1 Model Solution with scoring rubrics and step breakdown.
- Complete worked steps verified by university external examiners.